Spatial Intelligence World Model
空间智能世界模型

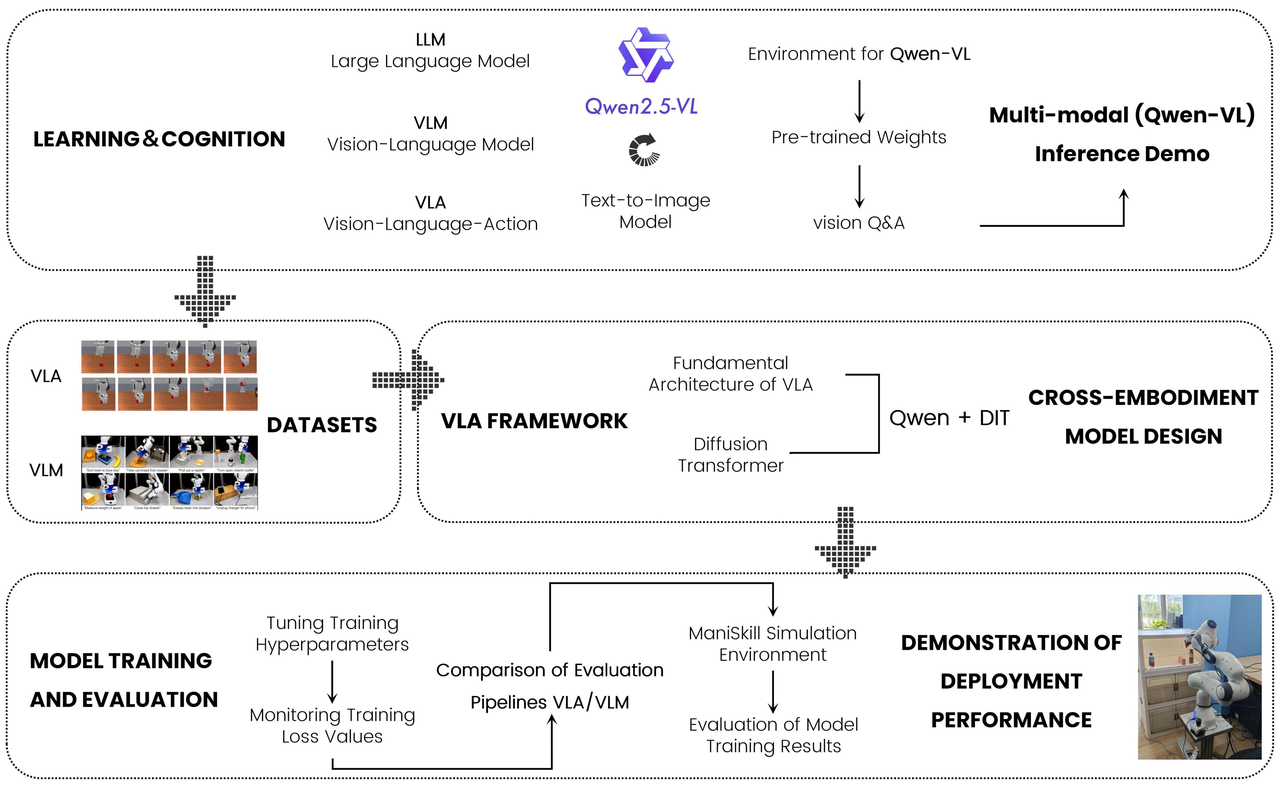
ROBOTIC SPATIAL INTELLIGENCE: WORLD MODELS FOR AUTONOMOUS MANIPULATION

INSTRUCTORS 导师：
Heng Tao Shen 申恒涛 | Lei Zhu 朱磊

TEACHING ASSISTANTS 助教：
Tianshi Wang 王天时 | Wentao Tan 谭文韬 | Heng Zhi 陟恒 | Chenyu Liu 刘晨宇

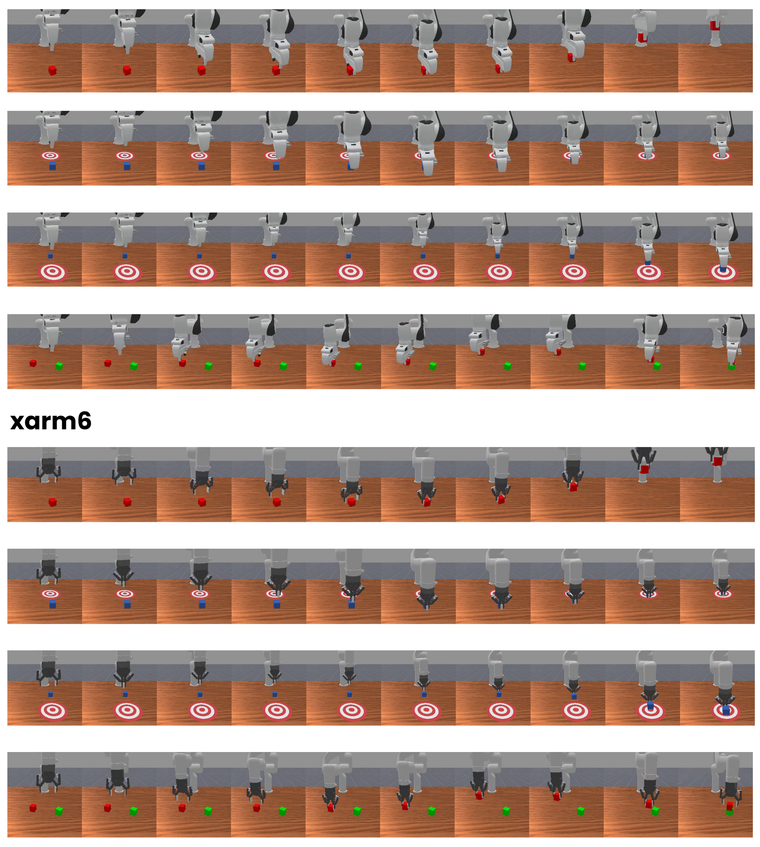
STUDENTS 学员：
Anji Wang 王安济 , Yang Yue 岳洋 , Yilun Jiang 蒋逸伦 , Zhipeng Guo 郭志鹏 , Hanwen Yu 于翰文 , Shilong Wei 魏仕龙 , Ruihan Gui 归瑞涵 , Min Wang 王敏

WORKFLOW



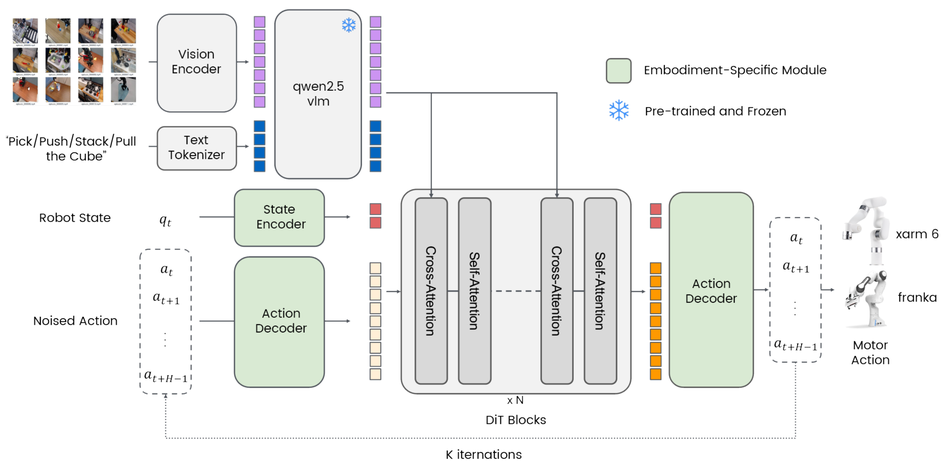
DATA TRAINING SAMPLE

franka



xarm6

METHODOLOGY



EQUIPMENT INTRODUCTION

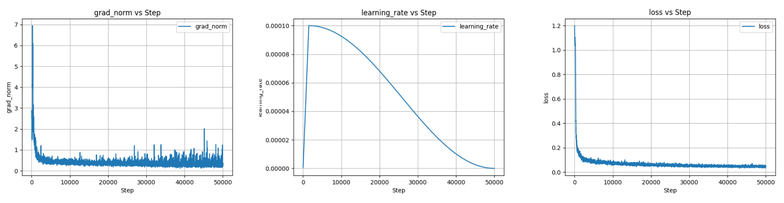
Franka 7-Axis Robotic Arm

The Franka 7-axis robotic arm (e.g., FR3/FP3 models) is a high-precision force-controlled cobot developed by Germany's Franka Robotics, now part of Agile Robots Group. Its 7-DOF design mimics human arm movements for complex tasks in confined spaces, while integrated torque sensors enable 0.05N force resolution with real-time feedback. Certified under ISO safety standards, it allows fence-free human collaboration and supports FCI, ROS, and MATLAB for research and industrial automation. With an 18kg lightweight structure, ±0.1mm repeatability, and 2m/s end-effector speed, it excels in medical, precision manufacturing, and AI training applications.

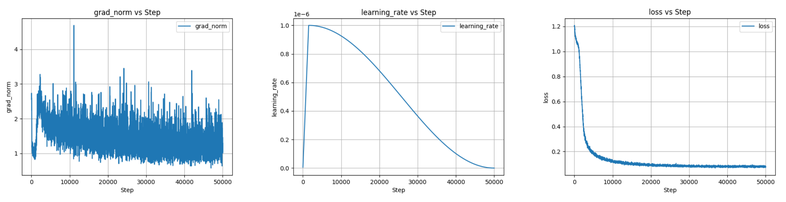
xArm6 Robotic Arm

The xArm6 is a cost-effective 6-axis collaborative robot by UFACTORY (Shenzhen), designed for education, research, and light industrial use. Featuring 6-DOF flexibility, 700mm reach, and ±0.1mm repeatability, it handles 5kg payloads for tasks like assembly and material handling. Compatible with ROS for simulation (Gazebo) and motion planning (MoveIt!), it enables rapid prototyping, while its safety design permits human interaction without barriers. User-friendly Python APIs and debugging tools simplify programming, making it ideal for academic labs, flexible production lines, and unmanned service applications.

TRAINING RESULTS



EXPERMENT 1



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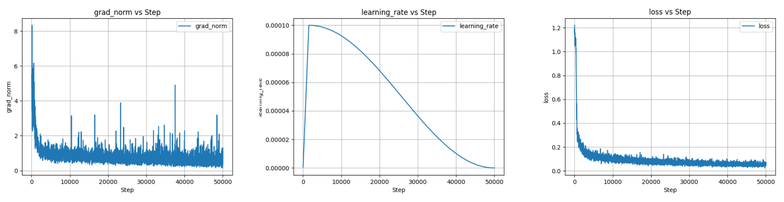

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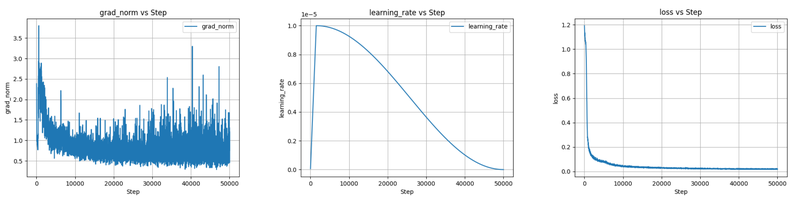

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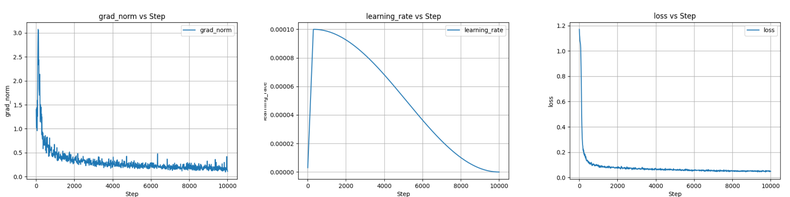

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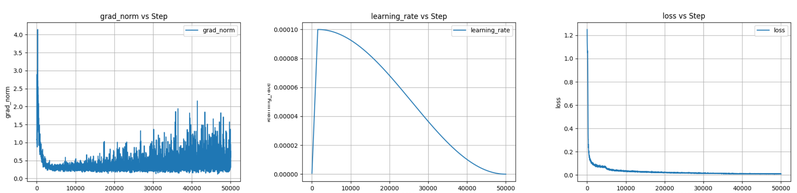
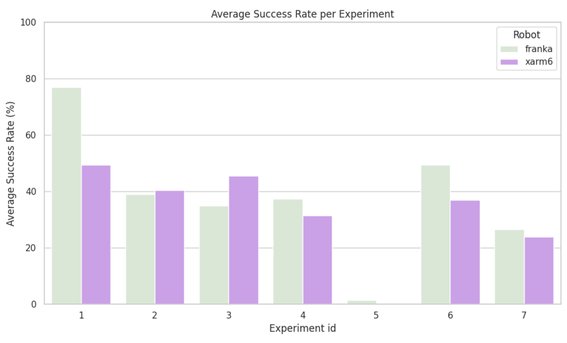
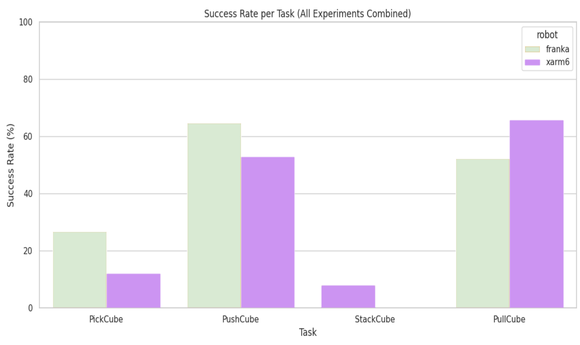
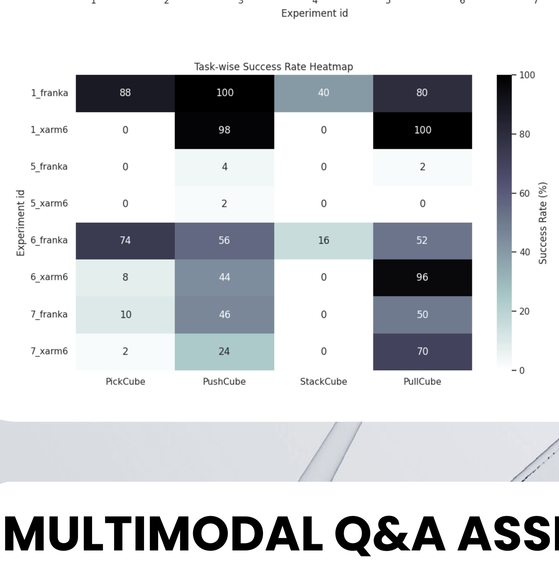


EXPERMENT 6

EVALUATION RESULTS







CONCLUTION

Franka is more sensitive to larger batch sizes, while the performance of xArm6 remains relatively stable across different batch sizes but shows no significant improvement.

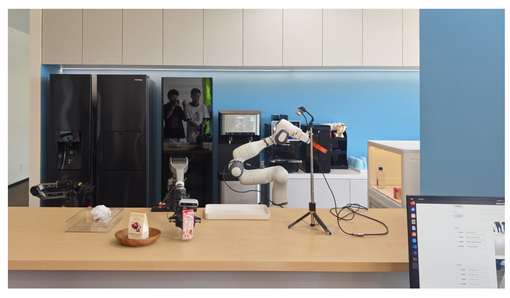
The number of training steps significantly impacts performance. Shorter training durations greatly reduce performance, indicating a lack of convergence.

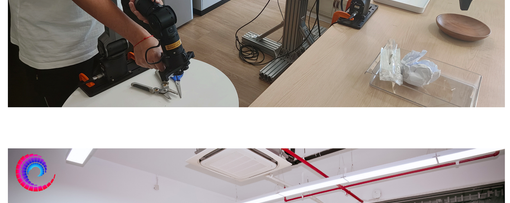
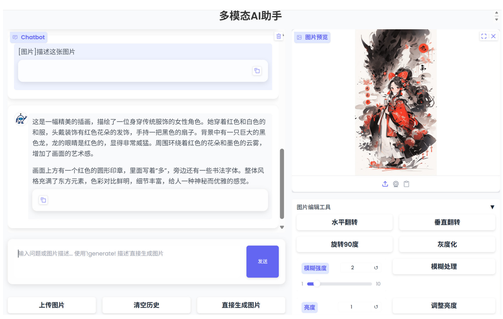
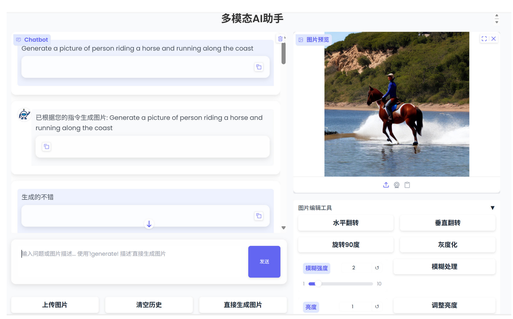
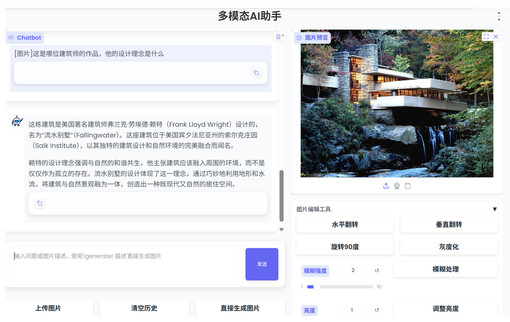
A learning rate of 1e-4 yields the best performance; 1e-2 is too large and unusable; 1e-6 is too slow and leads to degraded performance.

Different robots exhibit preferences for different tasks; StackCube performs poorly overall and requires separate tuning.

Experiments demonstrate that our framework is capable of generating actions across heterogeneous robotic arms.

PHOTOS





MULTIMODAL Q&A ASSISTANT